User Guide

Vela IF820 USB Dongle

Part # 450-00185

Version 2.1

Revision History

Version	Date	Notes	Contributor(s)	Approver
1.0	30 November 2023	Initial version	Erik Lins	Jonathan Kaye
2.0	15 May 2024	Converted to Ezurio formatting. Fixes for latest firmware.	Mark Duncombe	Jonathan Kaye
2.1	29 May 2024	Added section on updating the DVK Probe firmware. Updated section on HCI firmware.	Erik Lins Mark Duncombe	Jonathan Kaye

Contents

1 Overview

The Vela IF820 USB dongle (Ezurio part # 450-00185) is a packaged USB Adapter version of the integrated antenna Vela IF820 module. It uses a virtual serial port (VSP) implementation to enable full Bluetooth 5 operation into the widest range of Operating System backed devices with a USB interface.

The Vela IF820 series of BLE modules feature:

- Bluetooth BR / EDR & Bluetooth LE v5.4
- Based on Infineon AIROC™ CYW20820 chipset
- Bluetooth Low Energy
  - Support - Peripheral/Central roles
  - Support for 1 Mbps and 2 Mbps PHY
- Classic Bluetooth profiles
  - EZ-Serial (SPP only)
  - HCI UART or Modus Toolbox (any supported by respective Bluetooth SW stack)
- Powerful Core Cortex-M4, 96MHz: 256 kB on chip Flash, 176kB on-chip RAM, 1MB on chip ROM
- Industrial Temperature Range (-40°C to +85°C)

More information regarding this product series, including a detailed module data sheet and application notes, are available on the Vela IF820 product page.

1.1 Part Numbers

Part Number	Product Description
450-00185	USB dongle containing 453-00171 Vela IF820 module (Integrated antenna)

2 Vela IF820 USB Dongle

This section describes the Vela IF820 USB dongle hardware. The Vela IF820 USB dongle is delivered with the Vela IF820 series module loaded with Infineon's EZ-Serial firmware. It starts up in EZ-Serial mode by default.

The USB dongle allows the Vela IF820 series module to physically connect to a PC via a USB port which provides USB-to-Virtual COM port conversion. Any Windows PC (XP or later), Linux PC (Kernel 3.x or newer with an x86, x86_64 or ARMv7 CPU) or Mac (10.11 or newer), should auto-install the necessary drivers.

2.1 Key Features

The Vela IF820 USB dongle has the following features:

- Vela IF820 series module soldered onto the board
- Self-powered by USB port, running the module at a regulated 3.3v
- USB to UART bridge based on RP2040
- Vela IF820 UART can be interfaced to a PC via USB using the USB-UART bridge
- One LED for status indication
- Pinhole push-button
- EZ-Serial firmware via UART (using the Virtual COM port)
- application firmware upgrade capability via UART (bootloader using the Virtual COM port)

3 Understanding the Dongle

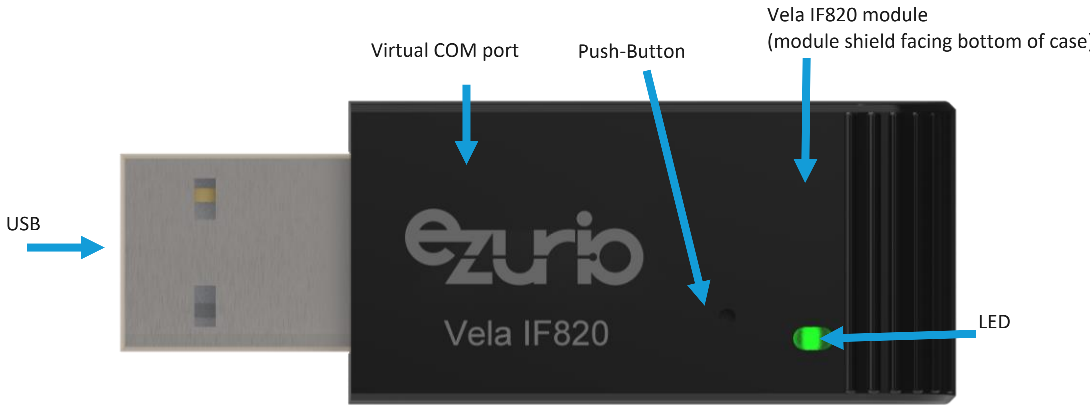


Figure 1: Vela IF820 dongle

3.1 UART Serial Interfaces

The Vela IF820 module exposes two UART interfaces which are accessible as Virtual COM ports on a PC.

3.1.1 EZ-Serial UART

The EZ-Serial UART is a four-wire UART interface (TX, RX, CTS, RTS) through USB as a Virtual COM port. The EZ-Serial UART connection on the IF820 module is shown below.

Table 1: EZ-Serial GPIO/UART connections

IF820 Pin	Default Function
25	BT_UART_TX (output)
26	BT_UART_RX (input)
24	BT_UART_RTS (output)
27	BT_UART_CTS (input)

3.1.2 HCI UART

The HCI UART is another four-wire UART interface (TX, RX, CTS, RTS) through USB as a Virtual COM port. See below for the HCI UART connection on the IF820 module.

Table 2: HCI GPIO/UART connections

IF820 Pin	Default Function
17 (P32)	HCI_UART_TX (output)
15 (P37)	HCI_UART_RX (input)
16 (P11)	HCI_UART_RTS (output)
14 (P10)	HCI_UART_CTS (input)

3.2 Identifying the Virtual COM Ports

The COM port assigned by the PC is arbitrary and both the numbers as well as the order of Virtual COM ports can differ from PC to PC. They can also depend on what other USB devices with Virtual COM ports are already connected.

For identifying which Virtual COM port number corresponds to the EZ-Serial UART or the HCI UART, it's necessary to open the Windows Device Manager (press Windows key and start typing “Device Manager” and the app should appear on top of the result list). You should see two entries for serial USB device with COM port numbers. Double click on each device, switch to the Details tab and select the “Bus reported device description” property. One of the COM ports will show as UART0, the other as UART1. The below table shows the default functions of the two ports.

Table 3: Virtual COM port numbering

COM port name	Default Function
UART0	IF820 HCI UART port
UART1	IF820 EZ-Serial UART port

Below is a screenshot of the Windows Device Manager showing the above process.

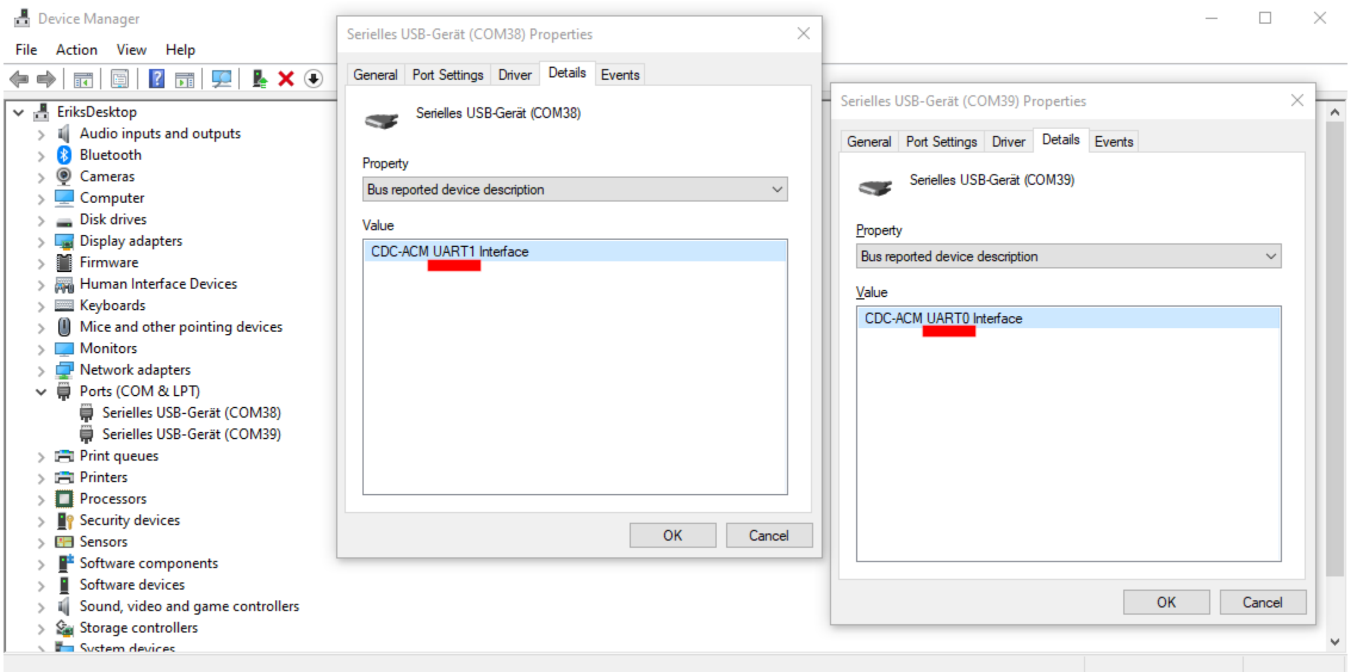


Figure 2: UART ports identified in Windows Device Manager

3.3 Additional Peripherals

3.3.1 LED

The Vela IF820 USB dongle has one integrated LED, the details of pin mapping are listed below.

Table 4: LED to IF80 pin mapping

Part	IF820 Pin	Comments
LED1	33 (P27)	Set low to turn LED on, high to turn off

3.3.2 Push-Button

A small push-button underneath a pinhole is connected directly to the RP2040 USB UART bridge controller as BOOTSEL pin. It’s not needed for regular EZ-Serial or HCI firmware update, but is provided in case the RP2040 firmware must be updated.

4 Operating Modes

Always make sure to have the latest DVK Probe firmware loaded to the RP2040 in the dongle. See section 5 for updating the DVK Probe firmware on the dongle.

4.1 EZ-Serial

EZ-Serial is a firmware platform built on top of the Vela IF820 Module, providing an easy-to-use method for accessing the most common hardware and communication features for dual-mode Bluetooth® applications. It uses a UART-based API for use with an external host MCU. Ezurio provides open-source Python host samples to test common use cases.

EZ-Serial supports cable replacement capability for both Bluetooth® Classic and Low Energy.

See our EZ Serial User Guide for the Vela IF820 Module for more information on EZ Serial and a command reference:

https://www.ezurio.com/documentation/user-guide-ez-serial-vela-if820-series

More information on using the EZ-Serial Firmware is available here:

https://lairdcp.github.io/guides/vela_if820_user_guide/latest

4.1.1 Determining the Firmware Version

Use the “system_query_firmware_version” /qfv or “system_reboot” /rbt commands to determine the EZ-Serial Firmware Revision. Both commands are illustrated below using a terminal program connected to the PUART.

```
/qfv
@R,002C,/QFV,0000,E=01040C0C,S=03010000,P=0104,H=F1
```

The E field is the EZ-Serial Firmware Revision. In this case it is 01040C0C in hex. This equates to 1.4.12.12

```
/rbt
@R,000A,/RBT,0000
@E,0076,BOOT,E=01040C0C,S=03010000,P=0104,H=F1,C=00,A=FF247F484E50,F=EZ-Serial-VELA_IF820_INT V1.4.12.12 Sep 11
2023 10:17:21
```

The F field is the EZ-Serial Firmware Revision string. This is a more friendly representation of the firmware version.

4.1.2 Updating the EZ-Serial Firmware

The latest EZ-Serial firmware release is available here:

https://github.com/LairdCP/Vela_IF820_Firmware/releases

There exist two version of the EZ-Serial firmware, one for the IF820 module with external antenna (IF820_EXT_ANT…zip) and one for the integrated antenna variant (IF820_INT_ANT…zip). For updating the IF820 USB dongle the integrated antenna version is needed. The firmware ZIP archive contains several files and make sure to extract all of them into one place.

The releases page also show two software tools for updating the firmware:

- If820_flasher_cli.exe is a Windows command line utility
- If820_flasher_gui.exe is a Windows utility with graphical user interface

4.1.2.1 Updating with the GUI Utility

Download the above if820_flasher_gui.exe utility and start. It should identify connected IF820 dongles or DVKs automatically and show them in the Select Board list. Select the board and then hit Browse to select the firmware hex file you want to program. “Upgrade Firmware” will then do the firmware upgrade automatically.

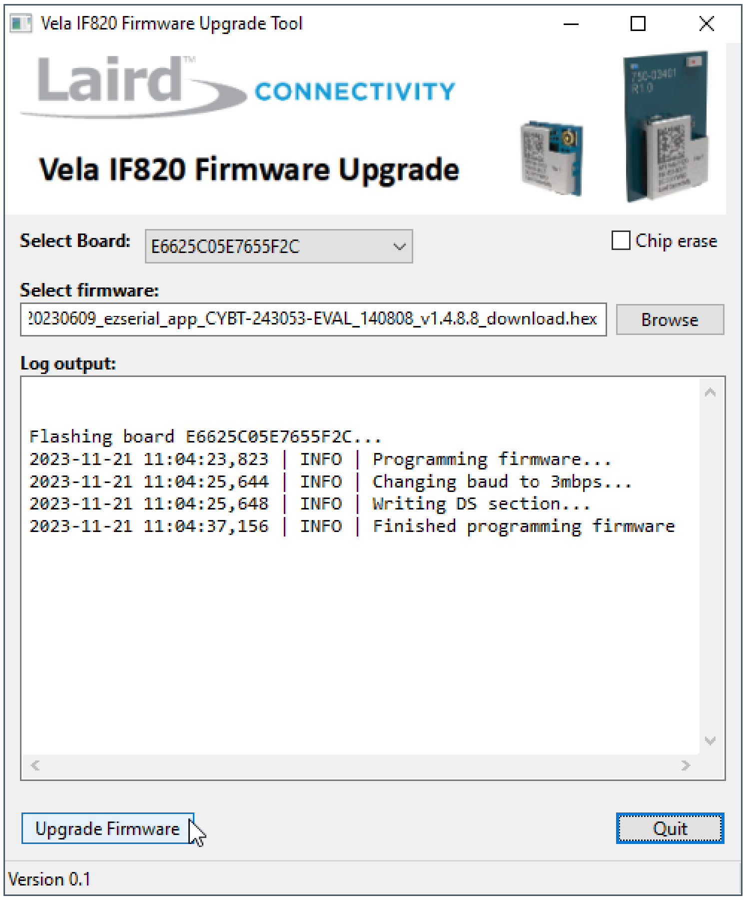


Figure 3: Vela IF820 firmware upgrade utility

4.1.2.2 Updating with the Command Line Utility

Download the above if820_flasher_cli.exe and open a command line window (Windows-key + R, type “cmd” and hit enter). Starting the utility with the parameter –help will show usage information with existing parameters.

The command line utility will identify connected dongles or DVKs automatically. If only one device is connected to the PC, there is no need to specifiy a COM port and the command would be:

```
If820_flasher_cli.exe -f 20230609_ezserial_app_CYBT-243053-EVAL_140808_v1.4.8.8_download.hex
```

The update with run through automatically as shown in Figure 4.

In the case of more than one dongle/board being connected to the PC, you must specify the proper HCI COM port.

```
C:\Windows\System32\cmd.exe

c:\Users\erikl\OneDrive - Laird Connectivity\work\02_BT_BLE_Modules\IF820\testing_binaries>if820_flasher_cli.exe --help
usage: if820_flasher_cli [-h] [-c CONNECTION] [-ce] [-d] [-f FILE]

CLI tool to flash an IF820 board (or compatible boards) with new firmware.
If no COM port is specified, the tool will automatically detect the board and flash it.
If there is more than one board detected, the user will be prompted to select the board to flash.
The CLI supports chip erase, firmware update, and flashing firmware with chip erase.

options:
  -h, --help            show this help message and exit
  -c CONNECTION, --connection CONNECTION
                        HCI COM port
  -ce, --chip_erase     perform full chip erase.
  -d, --debug           Enable verbose debug messages
  -f FILE, --file FILE  application hex file to flash

c:\Users\erikl\OneDrive - Laird Connectivity\work\02_BT_BLE_Modules\IF820\testing_binaries>if820_flasher_cli.exe -f 2023
0609_ezserial_app_CYBT-243053-EVAL_140808_v1.4.8.8_download.hex
2023-11-10 08:17:41,036 | INFO | Programming firmware...
2023-11-10 08:17:42,778 | INFO | Changing baud to 3mbps...
2023-11-10 08:17:42,781 | INFO | Writing DS section...
2023-11-10 08:17:53,398 | INFO | Finished programming firmware

c:\Users\erikl\OneDrive - Laird Connectivity\work\02_BT_BLE_Modules\IF820\testing_binaries>
```

Figure 4: if820_flaser_cli.exe

4.2 HCI

It is possible to program the Vela IF820 module with a specific HCI firmware version. The module can then be used as an HCI Bluetooth controller exposing an HCI UART interface.

More information on HCI development is available here:

https://lairdcp.github.io/guides/vela_if820_user_guide/latest

4.2.1 Obtaining the HCI Firmware

Download the Vela IF820 HCI firmware at the following link:

https://github.com/LairdCP/Vela_IF820_Firmware/releases

The HCI firmware (.hcd file extension) is included in the IF820_INT_ANT…zip archive under "HCI binaries" section (scroll down).

4.2.2 Programming the HCI Firmware

With the IF820 dongle plugged in to your PC. Flash the HCI firmware using Flasher CLI on a Windows PC and the HCD file internal antenna.

```
if820_flasher_cli_v2.0.0.exe -ce -f IF820_INT_ANT_HCI_download_v1.0.0.hcd
```

```
C:\Users\Mark.Duncombe\OneDrive - Laird Connectivity\Desktop\test folder>if820_flasher_cli_v2.0.0.exe -ce -f IF820_INT_ANT_HCI_download_v1.0.0.hcd
IF820 Flasher CLI v2.0.0
2024-05-17 12:35:15,752 | INFO | Loading minidriver...
2024-05-17 12:35:18,003 | INFO | Performing chip erase...
2024-05-17 12:35:18,018 | INFO | Chip erase finished
2024-05-17 12:35:18,018 | INFO | Changing baud to 3000000
2024-05-17 12:35:18,037 | INFO | Programming HCD file...
2024-05-17 12:35:21,137 | INFO | Finished programming!
```

The IF820 dongle will now be programmed with the HCI firmware.

4.2.3 Using the IF820 HCI Dongle on Linux

Linux usually comes with all necessary tools preinstalled. Those are:

- BlueZ (Official Linux Bluetooth/LE stack)
- `btattach` and `bluetoothctl` (command line utilities for configuring HCI devices and doing basic tasks)

Connect the Vela IF820 to your Linux PC. The IF820 USB dongle version can be directly plugged into a USB port. When using an IF820 DVK this is simply done by connecting a USB cable.

Open a command shell/terminal on Linux and enter **`bluetoothctl list`**. This will return the existing HCI devices. If the PC has an integrated Bluetooth/LE controller, this will be shown as default device.

Attach the IF820 module to the Linux HCI subsystem by setting the proper serial port, baud rate and other parameters.

Note: The `btattach` command will not return. It will continue to run. Open a new terminal to issue other commands.

```
sudo btattach -B /dev/ttyACM0 -P h4 -S 3000000
```

Enter `bluetoothctl` to interact with the bluetooth subsystem. The `list` command will show the available bluetooth controllers. The IF820 module will now be the default device.

```
bluetoothctl

[bluetooth]# list
Controller 00:25:CA:B2:06:89 uby #2 [default]
Controller E4:A4:71:47:E6:6C uby
```

Scan for BLE devices with the `scan le` command. The `devices` command will show the discovered devices.

```
[bluetooth]# scan le
[bluetooth]# SetDiscoveryFilter success
[bluetooth]# hci1 type 6 discovering on
[bluetooth]# Discovery started
[bluetooth]# [CHG] Controller 00:25:CA:B2:06:89 Discovering: yes
[bluetooth]# [NEW] Device 20:6D:31:FB:06:49 Firewalla Gold
[bluetooth]# [NEW] Device 6F:EF:EF:C3:7C:30 6F-EF-EF-C3-7C-30
[bluetooth]# [NEW] Device 1A:0D:AB:3F:96:30 1A-0D-AB-3F-96-30
[bluetooth]# [NEW] Device 73:BA:C5:5A:76:C1 73-BA-C5-5A-76-C1
[bluetooth]# [NEW] Device 55:CA:44:0A:97:3B 55-CA-44-0A-97-3B
[bluetooth]# [NEW] Device 60:25:49:6C:67:25 60-25-49-6C-67-25
[bluetooth]# [NEW] Device 75:C3:D8:54:AB:F3 75-C3-D8-54-AB-F3
[bluetooth]# [NEW] Device 56:58:5F:C0:E1:3F 56-58-5F-C0-E1-3F
[bluetooth]# [NEW] Device 63:33:38:93:2F:45 63-33-38-93-2F-45
[bluetooth]# [NEW] Device 47:8E:99:DC:D3:74 47-8E-99-DC-D3-74
[bluetooth]# [NEW] Device 4A:B6:E8:8C:3E:B6 4A-B6-E8-8C-3E-B6
[bluetooth]# [CHG] Device 6F:EF:EF:C3:7C:30 RSSI: 0xffffffc4 (-60)
```

4.3 Troubleshooting

The following may help when troubleshooting issues.

If you see the following, ensure the IF820 flasher CLI and firmware image are in the same folder, and you are running the IF820 flasher CLI from that folder.

```
C:\Users\Mark.Duncombe\OneDrive - Laird Connectivity\Desktop\test folder>if820_flasher_cli.exe -ce -f IF820_INT_ANT_HCI_download_v1.0.0.hcd
'if820_flasher_cli.exe' is not recognized as an internal or external command,
operable program or batch file.
```

If you see the following, check the IF820 DVK or Dongle is plugged in.

```
C:\Users\Mark.Duncombe\OneDrive - Laird Connectivity\Desktop\test folder>if820_flasher_cli_v2.0.0.exe -ce -f IF820_INT_ANT_HCI_download_v1.0.0.hcd
IF820 Flasher CLI v2.0.0
2024-05-17 12:42:37,422 | ERROR | No boards found
Traceback (most recent call last):
  File "\\Mac\Home\git\vela_if820\Vela_IF820_Firmware\if820_flasher_cli.py", line 84, in <module>
NameError: name 'exit' is not defined
[25416] Failed to execute script 'if820_flasher_cli' due to unhandled exception!
```

You can access the help screen for the IF820 flasher CLI by using the -h option as below.

```
C:\Users\Mark.Duncombe\OneDrive - Laird Connectivity\Desktop\test folder>if820_flasher_cli_v2.0.0.exe -h
usage: if820_flasher_cli [-h] [-c CONNECTION] [-ce] [-d] [-f FILE] [-v]

CLI tool to flash an IF820 board (or compatible boards) with new firmware.
If no COM port is specified, the tool will automatically detect the board and flash it.
If there is more than one board detected, the user will be prompted to select the board to flash.
The CLI supports chip erase, firmware update, and flashing firmware with chip erase.

options:
  -h, --help            show this help message and exit
  -c CONNECTION, --connection CONNECTION
                        HCI COM port
  -ce, --chip_erase     perform full chip erase.
  -d, --debug           Enable verbose debug messages
  -f FILE, --file FILE  application hex file to flash
  -v, --version         Print the version of the tool and exit.
```

PyOCD (https://pyocd.io/docs/installing) can be used to check the DVK Probe firmware version using this command.

```
pyocd list -v -v -v -v
```

```
C:\Users\Mark.Duncombe\OneDrive - Laird Connectivity\Desktop\test folder>pyocd list -v -v -v -v
0001133 D Project directory: C:\Users\Mark.Duncombe\OneDrive - Laird Connectivity\Desktop\test folder [session]
0006861 D CMSIS-DAP v2 probe E6625C05E789602C: firmware version 2.0.0+1715182323, protocol version 2.1.1 [dap_access_cms
is_dap]
  #   Probe/Board                    Unique ID          Target
-------------------------------------------------------------------
  0   Ezurio DVK Probe CMSIS-DAP     E6625C05E789602C   ✔ cortex_m
      Ezurio                         Module
```

4.4 Custom Firmware Development

The Vela IF820 is fully supported by the Infineon ModusToolbox™. ModusToolbox™ releases provide the latest ROM patches, drivers, and sample applications allowing customized applications using the CYW20820 to be built quickly and efficiently.

More information on Custom Firmware Development is available here:

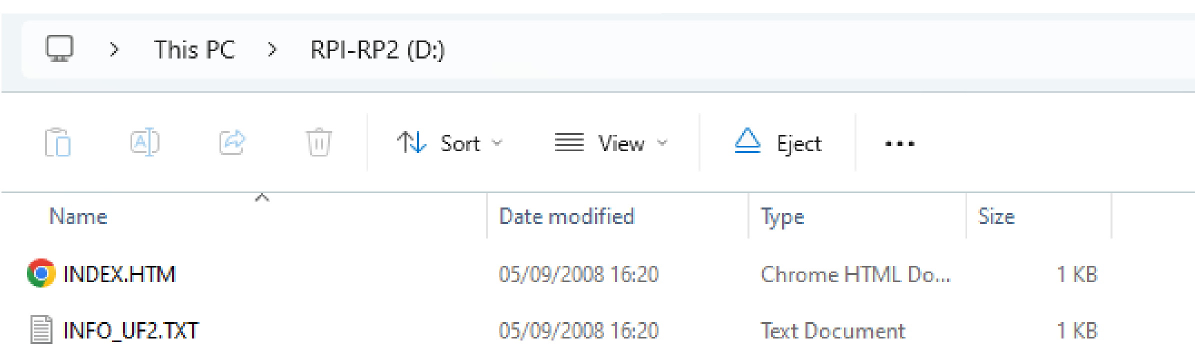
https://lairdcp.github.io/guides/vela_if820_user_guide/latest

5 DVK Probe Firmware

Always make sure to have the latest DVK Probe firmware loaded to the RP2040 in the dongle.

5.1 Updating over USB Mass Storage

Flash the DVK Probe firmware on a IF820 dongle by using a pin to press the boot select while plugging the dongle in. For the IF820 DVK, press and hold the boot select button while plugging the DVK in. This will make the DVK Probe appear as a drive in Windows.



Drag and drop the picoprobe.uf2 file from the firmware download to this drive, which will cause the dongle to reboot with the new RP2040 firmware.

6 Additional Documentation

Ezurio offers a variety of documentation and ancillary information to support our customers through the initial evaluation process and ultimately into mass production. Additional documentation can be accessed from the Documentation tab of the Vela IF820 Product Page.

7 Additional Information

Please contact your local sales representative or our support team for further assistance:

Headquarters	Ezurio 50 S. Main St. Suite 1100 Akron, OH 44308 USA
Website	http://www.ezurio.com
Technical Support	http://www.ezurio.com/resources/support
Sales Contact	http://www.ezurio.com/contact

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